Executive Summary

The present study is about analyzing the factors that can help the Vietnam government in motivating its citizens to get a regular health checkup. Factors such as demography, cost, health quality, awareness towards health and attitude play a major role when citizen decides to get a regular checkup. Due to budget constraints, the government wants to find which factor they should focus on while designing awareness campaigns that show the most improvement.

Since in the given data large number of survey variables are available with both numeric and category type, random forest models are more suitable. These models can be used to find the importance of variables.

Based on exploratory data analysis we chose certain variables for our random forest models. Using cross-validation we chose number of tress in random forestas 3 and the number of random parameters as 10.

We have verified our model using cross-validation. The test accuracy for the model was 30.32% as shown in Table 0.1

Predicted	**Actual**		**Accuracy in percentage**
	No	**Yes**	
No	101	51	30.32%
Yes	43	115	

Table 0.1: Confusion matrix for verifying random forest model on test set. Classify people according to whether they go for regular checkup. Both False positive and False negative error are around 30%

The model suggested that variables Job-status, body mass index and reliability on health service were most important factor while classifying people who opt for regular checkup.

In general affordability , awareness , Habit and reliability towards public health service was most important. Some people were willing to go for regular checkup if they were informed by mobile applications.

Further it was found that people of age group 18-29 with graduate-level education and people with a stable income can be convinced most easily. Hence, the government should design their their campaign based on these groups of people

Analyzing factors affecting the choice of regular checkup in Vietnam

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1 Introduction

Countries without established socialized medical systems can impose serious hardships for its citizens especially from a poorer background. Even in a developed country like the USA treatment of serious conditions are very expensive. Ideally, patients would get regular check-ups (or "general health examinations", GHEs) so that serious conditions could be detected and treated early before they cause serious problems. This would save money and improve public health.

However, many people avoid general health examinations due to various reasons such as they might be too expensive, or too difficult to schedule; some people may not trust doctors or believe that check-ups have any value, or some people may have had bad experiences when they previously tried to get check-ups. Even due to lack of sufficient information many groups such as the anti-vaccine group spread disinformation leading to fear and no trust in doctors and basic medicines. Hence, the government must provide affordable health care as well as create awareness about regular checkups so that a serious epidemic in society can be avoided.

For this reason, public health researchers in Vietnam wanted to explore reasons that prevented widespread use of regular check-ups. they want to know why people do not sign up for exams and understanding why could help them make policy changes or make their public relations campaigns more effective. They could simply send reminders or offer incentives to anyone who does not appear for an exam each year. Since the budget is limited they cannot afford to try to convince every possible patient to get a check-up. Hence they are interested in knowing what factors affect most of its citizens to avoid regular checkups. They can then create awareness program based on those important factors. For example, if people tend to think check-ups are a waste of money, their advertising should emphasize how cheap they are; if people don't believe medical service is high-quality, the advertising should emphasize the expertise and training of doctors.

They conducted an interview survey in Hanoi and Hung Yen, Vietnam. They interviewed people in person for about 10-15 minutes at secondary schools, hospitals, companies, government agencies and randomly selected households.

In the present study using this survey we try to understand what factors prevent citizens from opting for regular checkups.

2 Data Summary

We obtained data set from the response of surveys conducted by Vietnamese government and provided in Q-H Vuong with the description of variables. It contains a response from 2,068 different citizens. Given data does not contain any missing values hence no sample was removed.

As explained in section 2.1 we label each sample as 0 and 1, where 1 represents citizens opting for regular checkup. We then explore factors that affect most in the classification.

2.1 Defining labels for each samples

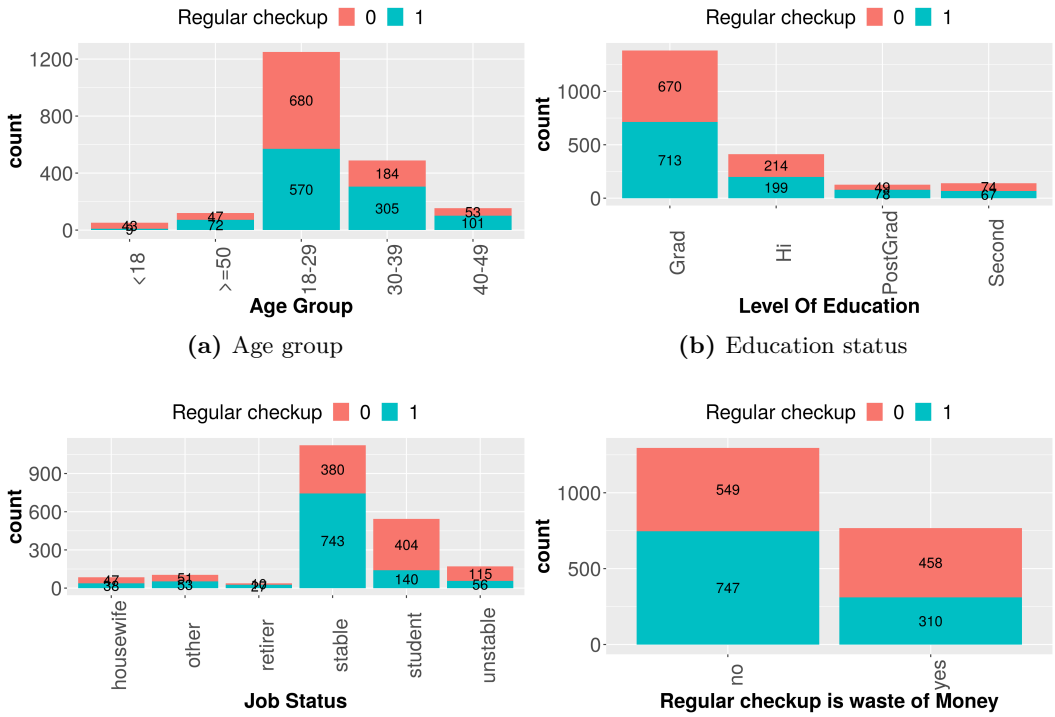
For the present study, we have considered regular checkups when the respondent last went for check-up less than 12 months. In the given raw data, we use variable "RecPerExam" that describes the time since the respondent last visited a doctor for a check-up, not prompted by a specific illness. Hence, this variable indicates how much a respondent gives importance to regular checkups. This variable can take 4 values, "**less12**" : less than 12 months, "**b1224**" : between 12 and 24 months, "**g24**" : over 24 months and "**unknown**" : respondent doesn't know . Hence for any respondent whose "RecPerExam" value is "less12" we label them as **1** else **0**.

3 Exploratory Data Analysis

3.1 Handling outliers

There are 4 respondents whose age is less than 16. Since the decision to get a regular checkup of these respondents depends on their guardians and not based on what they feel, we have considered them as outliers and removed from the data.

3.2 Summary of factors affecting regular checkup from the survey



(a) Age group

(b) Education status

(c) Job Status

(d) Regular checkup is waste of money 4

Figure 3.1: From fig.(c) and fig.(d) financial status plays important role in deciding not go for regular checkup. Apart from that people with more education go for regular checkup whereas most young people do not go for regular checkup

We have grouped variables from the given data into 6 different categories and discussed how these categories affects citizen's decision for opting taking regular checkups.

Finance

It has been found that financial condition plays a major role in motivating citizens to opt for regular checkup. From the survey around 60% of the respondents who are receiving permanent treatment and around 60% whose family member is receiving long term treatment tends to avoid regular checkup similarly. Indicating that long term treatments might lead to financial hardship therefore they avoid regular checkups. Around 50% respondents who do not go for regular checkup feels that checkup should cost less than 2 million VND. Again indicating cost of checkup

playing a major role. And, if provided cash, around 60% suggests that they will prefer going soon. 70% respondents with unstable income , 75% of respondent with occupation as students and 55% housewives do not go for a regular checkup while more than 70% of respondents with who are retired or have stable income go for regular checkups, indicating the financial stability plays important role in this decision. Finally, Around 60% of respondents with insurance or company subsidized checkups go for regular checkup.

Perceived quality of health service

Perceived quality of service is found to have major role here in decisions of opting for regular checkups. Overall most people rate the quality of public health care as above average. But those who rated below average quality do not go for regular checkup. For example , around 40% have no faith in medical services. Of them 53% do not go for regular checkup. Around than 30% of all respondents rated timeliness of medical checkup as below average and around 54% of them do not go for regular checkup. In terms of perceived quality of medical equipment only 14% rated it as below average and 59% of them do not go for regular checkup. Regarding, perceived thoughtfulness and sense of responsibility of medical staff 23% rated below average and 60% of them do not go for regular checkup. Similarly, 14% rated the knowledge or ability to assure professionals as below average and 51% of them do not go for regular checkup. Finally, 17% of respondents rated perceived ability of examiner to perform medical services below average and 53% of them do not get regular checkups.

Awareness

Only 35% of respondents believe that general public health is bad and 55% of them go for regular checkup. For rest 65% of respondents who believe public health is good and quiet or do not know about public health, more than 50% of them do not opt for regular checkup indicating that health awareness does play role in opting for regular health checkup. Only 62% of respondents with higher education levels such as Ph.D. go for a regular checkup compared to 48% with just higher secondary or just middle school or graduate education. More than 53% of respondents who do not know how to use basic medical equipment such as blood pressure monitor and 60% of those who do not keep medicine cabinet at their home tend to avoid regular checkup. And 56% of respondents who do not follow health measures tend to avoid regular checkup. All these suggest lack of awareness affect decision of not getting regular checkup. Finally, the survey suggests that 75% of respondents are willing to use the healthcare app and more than of the 85% say that they will go for a checkup if the app suggests them. All these suggest that more awareness motivates for a regular checkup.

Habits

It has also been found that respondents who give health a priority, do a lot of exercises, constantly follow health measures and whose family have stable health go for a regular checkup. More respondents with high body bass index opt for a regular checkup.

Perception

Most respondents do believe than they should go for a regular checkup once a year. And around 50% of them who believe this go for the regular checkup. However, the reason for the other 50% of them not going for a regular checkup can be due to belief checkups are not urgent and important or they think it is a waste of money and time.

Demography

Most of the respondents lie between the age group of 18 and 29. But very few of them go for a regular checkup. Respondents with age > 30 prefer going for a regular checkup. Out of them, married go for a regular checkup. Males are slightly more inclined towards regular checkups than females. The place doesn't seem to affect the motivation for getting a regular checkup.

3.3 Observed relationship among variables

Variable weight is highly correlated with body mass index and height also height is highly correlated with gender. Similarly, marital status is related to job status and Age group. Age and Age group are highly correlated. Hence we decided to drop variable height, weight, marital status and age from our model.

Variables tangible, reliability, response, assurance and empathy are highly correlated among each other hence we decided to drop all except reliability. Similarly, variables sufficiency of information, The attractiveness of information, the impressiveness of information and the popularity of information are highly correlated. Hence we decided to consider only the popularity of information in our model.

4 Method

Based on exploratory data analysis, using variables which were not dropped we built our random forest model to classify whether respondent prefer regular checkup or not. We have divided our 2068 samples into 1754 training and 310 test set. Then

using cross-validation we estimated the optimum value of hyper-parameters that is the number of trees (**ntree**) and the number of variables to be chosen randomly (**nmtry**). As a result for best validation error, we obtain the value of ntree as 3 and nmtry as 10

With obtained ntree and nmtry and using the complete training set we train our random forest model

4.1 Model Diagnostic

Name	Fold1	Fold2	Fold3	Fold4	Fold5	Fold6	Fold7	Fold8	Fold9	Fold10
Acc.	0.267	0.2629	0.3143	0.2443	0.2971	0.2857	0.2557	0.2514	0.2457	0.2216

Table 4.1: Cross validation for each fold. Accuracy does vary, suggesting model trained suffer from some sample bias

Predicted	**Actual**		**Accuracy in percentage**
	No	**Yes**	
No	101	51	30.32%
Yes	43	115	

Table 4.2: Confusion matrix for verifying random forest model on test set. Classify people according to whether they go for regular checkup. Both False positive and False negative error are around 30%

Table 4.1 shows that cross-validation error is consistent accuracy does not vary much with the training set. It has an average error of approx .25%. This error is close to almost with test error which is 30.32% as shown in table 4.2.

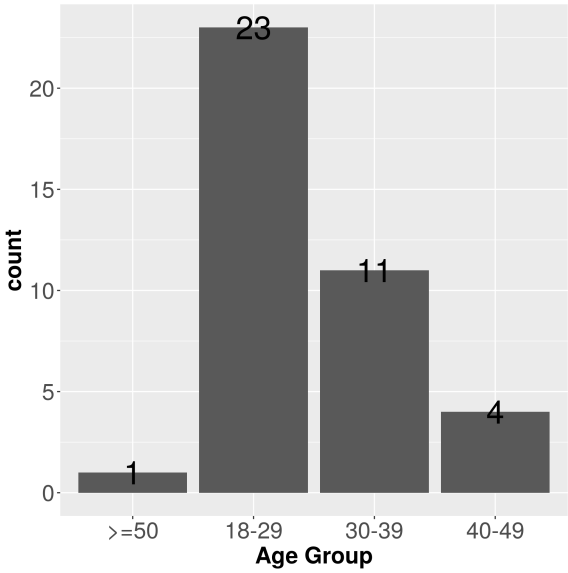
4.2 Model Inference

Table 4.3: Variable importance for each fold. Suggest job status and Body mass index most important variable

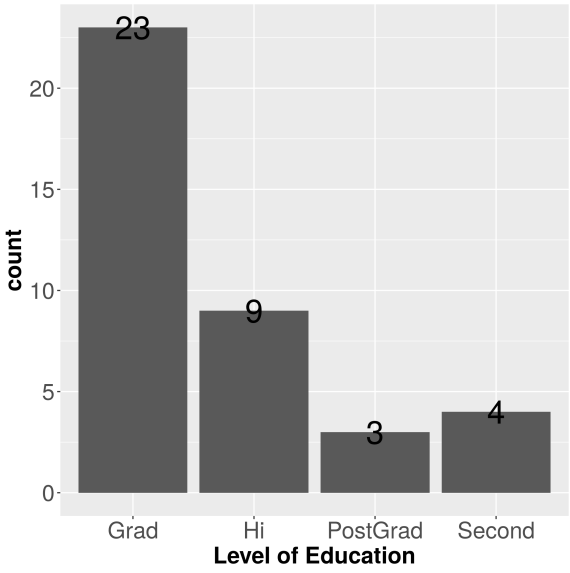
Variable	Importance
Job status	79.17
Education	25.59
place	25.65
Body Mass Index	88.30
Health Insurance	20.14
Regular checkup Waste of time	21.35
Regular checkup Waste of money	18.95
Regular checkup Not Important	32.21
Habit	42.01
Follow Health update	22.49
Receive permanent treatment	15.42
Keep Medicine cabinet	10.68
Reliable	17.38
Empathy	39.09
Affordable Cost	38.46
Use IT app	26.96
Regular checkup using IT app	25.89
Age group	36.53

Table 4.3 suggest job status is the most important factor in deciding whether or not a person will opt for a regular checkup. Although perception about health service from variables Empathy, Reliable and Affordable cost does play a significant role.

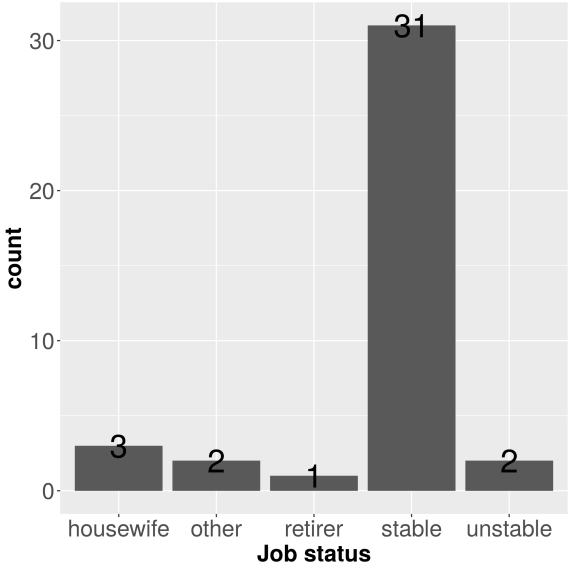
We studied respondents who do not go for a regular checkup but our model predicts them that they for a regular checkup. These people are easier to convince. Base on figure 4.1 government should target people between the age group of 18-29 with graduation degree or people with stable income. They will be most easy to convince



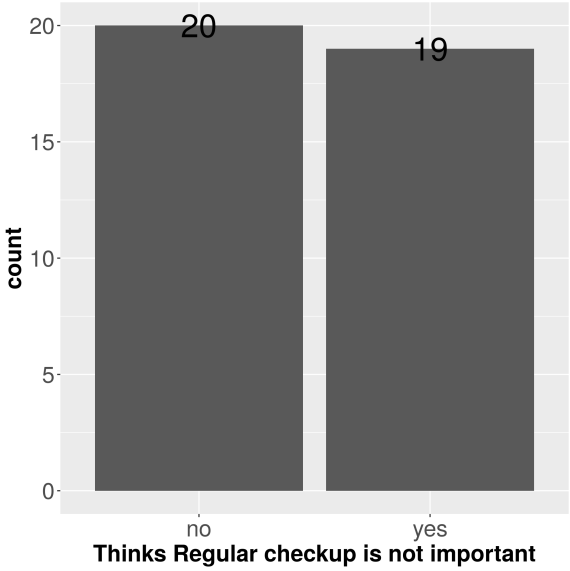
(a) Age group



(b) Education status



(c) Job Status



(d) Regular checkup not important

Figure 4.1: Respondents most easy to convince based on model. Most of the respondents do not think regular checkup is not important, hence people with stable income, between age group of 18-29 and with education of graduate level should be targeted

5 Conclusion

A study of factors affecting the person's ability to go for a regular checkup was carried out using a random forest model. The test accuracy for the model was 30.32%. It suggested that variables like Job-status, body mass index and reliability on health service were most important.

On studying the people who do not go for a regular checkup but our model classified them that they go, we found that people of age group 18-29 with graduate-level education and people with a stable income can be convinced most easily as they were very close to classification boundary.

The model still needs a certain improvement. The proper choice of a variable is important. Carefully discarding variables that are correlated to other variables is very important. In conclusion, the model does give some information about people who do not go for a regular checkup and can be used by the government to design their campaign

6 References

Q-H Vuong (2017). 'Survey data on Vietnamese propensity to attend periodic general health examinations. ' *Scientific Data 4 170142* https://doi.org/10.1038/sdata.2017.142.